

Lesson 18 Equivalence Relations

In Lesson 17, we gave a formal definition of relations and three properties a relation may have: reflexivity, symmetry, and transitivity. In this Lesson, we will focus on relations which possess all these three properties.

Definition A relation is an *equivalence relation* if it is reflexive, symmetric, and transitive.

We show that " $=$ " on $\mathbb{Z}$ is reflexive, symmetric, and transitive, so " $=$ " is an equivalence relation on $\mathbb{Z}$. There are many other equivalence relations on $\mathbb{Z}$, for example, the "**same parity**" relation is an equivalence relation.

- Reflexive: For every $x \in \mathbb{Z}$, xRx because x has the same parity as itself.
- Symmetric: Let $x, y \in \mathbb{Z}$. If xRy , then x has the same parity as y , which implies that y has the same parity as x , that is, yRx .
- Transitive: Let $x, y, z \in \mathbb{Z}$. If xRy and xRy , then x has the same parity as y , and y has the same parity as z . Hence x has the same parity as z , that is, xRz .

Equivalence relations extend the notion of "equality"; things which are "similar" are related or "equivalent" though they are not equal: $1 \neq 3$ but parity wise, 1 is equivalent to 3. We now recall an important definition (Lesson 1 Exercises).

Definition Let $n \in \mathbb{N}$. Two integers x and y are *congruent modulo n* , denoted $x \equiv y \pmod{n}$, if $n|(x - y)$ or $x - y = an$ for some integer a .

Proposition 1 "Congruence modulo n " on $\mathbb{Z}$ is an equivalence relation.

Proof Let R be the "congruence modulo n " relation. Let $x, y, z \in \mathbb{Z}$.

R is reflexive: $x - x = 0 = 0(n)$, so xRx .

R is symmetric: If xRy , then $x - y = an$ for some $a \in \mathbb{Z}$, and $y - x = -an = (-a)n$. Since $-a$ is an integer, yRx .

R is transitive: If xRy and yRz , then $x - y = an$ and $y - z = bn$ for some $a, b \in \mathbb{Z}$. So $x - z = (x - y) + (y - z) = an + bn = (a + b)n$, xRz as $a + b$ is an integer.

Definition Let R be an equivalence relation on a set A and let $a \in A$. The *equivalence class* of a is the set of all elements of A related to a ,

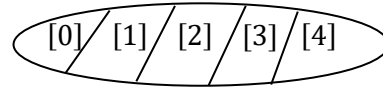
$$[a] = \{x \in A : xRa\}$$

Consider the "congruence modulo 5" relation on $\mathbb{Z}$. The equivalence classes are:

$$\begin{aligned} [a] &= \{x \in \mathbb{Z} : x - a = 5m, m \in \mathbb{Z}\} = \{5m + a : m \in \mathbb{Z}\} \\ [0] &= \{5m : m \in \mathbb{Z}\} = \{\dots, -5, 0, 5, 10, \dots\}, \quad [1] = \{5m + 1 : m \in \mathbb{Z}\} \\ &= \{\dots, -4, 1, 6, 11, \dots\} \\ [2] &= \{5m + 2 : m \in \mathbb{Z}\} = \{\dots, -3, 2, 7, \dots\}, \quad [3] = \{5m + 3 : m \in \mathbb{Z}\} \\ &= \{\dots, -2, 3, 8, \dots\} \\ [4] &= \{5m + 4 : m \in \mathbb{Z}\} = \{\dots, -1, 4, 9, \dots\} \end{aligned}$$

Note that $[5] = \{5m + 5 : m \in \mathbb{Z}\} = \{5(m + 1) : m \in \mathbb{Z}\} = [0]$, similarly $[6] = [1]$, $[7] = [2]$, etc. The “congruence modulo 5” relation divides $\mathbb{Z}$ into 5 “families”:

$$[k] = \{5m + k : m \in \mathbb{Z}\}, \quad k = 0, 1, 2, 3, 4$$



Integers with the same remainder are in the same “family”. In general, an equivalence relation on A collapses A into a set of non-overlapping families, as we see in the following theorem, which we will prove in Lesson 19.

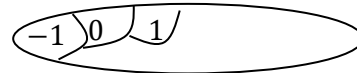
Theorem 2 Let R be an equivalence relation on a set A , and let $a, b \in A$.

- $a \in [a]$
- aRb if and only if $[a] = [b]$
- If $[a]$ and $[b]$ are not disjoint, then $[a] = [b]$.

The theorem says that every element of A belongs to an equivalence class, so A is divided into a set of nonempty equivalence classes whose union is A . Moreover, different equivalence classes are disjoint. Such a collection of disjoint nonempty subsets of A is a **partition** of A , and we have:

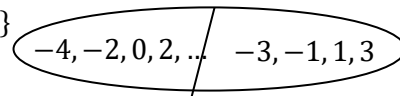
Corollary 3 An equivalence relation R on a set A defines a partition on A .

For the “equality” relation on $\mathbb{Z}$, the equivalence classes are “singletons” – every equivalence class has exactly one element, itself: $[a] = \{a\}$.



For the “same parity” relation on $\mathbb{Z}$, there are two distinct equivalence classes:

- the set of even integers: $[0] = \{\dots, -4, -2, 0, 2, 4, \dots\}$
- the set of odd integers: $[1] = \{\dots, -3, -1, 1, 3, \dots\}$



Consider the following relations on $A = \{1, 2, 3\}$.

- $R = \{(1,1), (2,2), (1,3), (3,1), (3,3)\}$ is an equivalence relation (can check for reflexivity, symmetry, and transitivity), with two equivalence classes:
 $[1] = [3] = \{1, 3\}, \quad [2] = \{2\}$
- $R = \{(1,1), (1,2), (2,1), (2,2), (2,3), (3,3)\}$ is NOT an equivalence relation, as it is not transitive: $1R2$ and $2R3$, but $1 \not R 3$.
- $R = \{(1,1), (1,2), (2,3), (3,1)\}$ is NOT an equivalence relation, because it is not reflexive: $2 \not R 2$.
- $R = A \times A$ is an equivalence relation, with only one equivalence class:
 $[1] = [2] = [3] = \{1, 2, 3\}$